

Exam Part 1: Mandatory Assignment 3

Exam number: 37 & 34

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Contributions. 34: Exercises 1, 3, 5. 37: Exercises 2, 4, 6.

Exercise 1: Investment Universe

We construct the investment universe from CRSP monthly data starting January 1980. A stock is included in month t if its absolute price exceeds \$2 *and* its lagged market capitalization exceeds the 20th percentile of NYSE-listed stocks in that month. The price screen eliminates penny stocks susceptible to bid-ask bounce, while the NYSE size percentile removes micro-cap stocks that are illiquid and poorly covered. Together, these filters correspond to the standard screens applied in the empirical asset pricing literature, following [Fama and French \(2008\)](#).

Table 1: Investment universe summary statistics (1980–2024).

Metric	Value
Full sample (1980–2024) – avg N	2007
Backtest period (2010–2024) – avg N	1898
Backtest period – min N	1726
Backtest period – max N	2319
Top 150 stocks – avg % of universe market cap	65.2%
Start date	1980-01-01
End date	2024-12-01

Table 1 summarizes the universe: on average 1898 stocks during the evaluation window (1726–2319 per month). Figure 1 shows the count is not monotonic: it rises from 1,723 stocks in 1980 to a peak of 2,924 in 2000 (the dot-com listing boom), contracts to 1,746 by 2011, then fluctuates, ending at 1,876 in 2024. For the optimization we cap each month’s universe at the top 150 stocks by market cap, still capturing 65% of filtered-universe market cap on average. This is a deliberate computational choice: with a fixed 60-month window, larger N adds parameters to $\hat{\mu}_t$ and $\hat{\Sigma}_t$ without adding observations, so the optimized portfolios estimate more poorly and trade more. We verify in Exercise 5 (Table 4) that the cap does not drive our conclusions.



Figure 1: Number of stocks in the investment universe per month, 1980–2024.

Exercise 2: Transaction-Cost-Adjusted Portfolio Optimization

2a. Closed-form solution. We form the Lagrangian

$$\mathcal{L} = \omega' \hat{\mu} - \frac{\gamma}{2} \omega' \hat{\Sigma} \omega - \lambda (\omega - \omega_{t+})' \hat{\Sigma} (\omega - \omega_{t+}) - \kappa (\iota' \omega - 1).$$

The first-order condition is

$$\hat{\mu} - \gamma \hat{\Sigma} \omega^* - 2\lambda \hat{\Sigma} (\omega^* - \omega_{t+}) - \kappa \iota = 0,$$

which rearranges to

$$(\gamma + 2\lambda) \hat{\Sigma} \omega^* = \hat{\mu} + 2\lambda \hat{\Sigma} \omega_{t+} - \kappa \iota,$$

giving

$$\omega^* = \underbrace{\frac{1}{\gamma + 2\lambda}}_{\equiv c} \hat{\Sigma}^{-1} (\hat{\mu} - \kappa \iota) + \frac{2\lambda}{\gamma + 2\lambda} \omega_{t+}.$$

Imposing $\iota' \omega^* = 1$ (and using $\iota' \omega_{t+} = 1$) yields

$$\kappa = \frac{\iota' \hat{\Sigma}^{-1} \hat{\mu} - \gamma}{\iota' \hat{\Sigma}^{-1} \iota},$$

which is *identical* to the Lagrange multiplier of the unconstrained fully-invested mean-variance portfolio ω^{MV} satisfying $\gamma \hat{\Sigma} \omega^{MV} = \hat{\mu} - \kappa \iota$. Substituting,

$$c \hat{\Sigma}^{-1} (\hat{\mu} - \kappa \iota) = \frac{\gamma}{\gamma + 2\lambda} \omega^{MV},$$

so the closed-form solution is

$$\omega_{t+1}^* = \frac{\gamma}{\gamma + 2\lambda} \omega^{MV} + \frac{2\lambda}{\gamma + 2\lambda} \omega_{t+}.$$

Since both coefficients are positive and sum to one, ω_{t+1}^* is a convex combination of the mean-variance optimal portfolio and the pre-rebalancing weight. The parameter λ governs the trade-off: $\lambda \rightarrow 0$ recovers ω^{MV} , $\lambda \rightarrow \infty$ leaves the portfolio unchanged at ω_{t+} .

2b. Realism of variance-proportional transaction costs. Modeling costs as $\lambda(\omega - \omega_{t+})' \hat{\Sigma}(\omega - \omega_{t+})$ makes rebalancing a stock more costly the higher its return variance. This is partially realistic, since volatile stocks tend to carry wider spreads and larger impact, but it has three weaknesses. First, real costs are closer to *linear* in trade size than quadratic. Second, the binding friction is order-book depth and volume, not variance (liquid large-cap tech is volatile yet cheap to trade, which this still penalizes). Third, the quadratic form is symmetric across purchases and sales. It is best read as an analytically convenient approximation, not a structural cost model.

2c. Implementation. We implement `compute_weights(mu, Sigma, gamma, lam, w_prev)` exactly as derived in 2a. Given $\hat{\mu}$, $\hat{\Sigma}$, γ , λ , ω_{t+} , the function: (1) adds a ridge term $\varepsilon = \max(10^{-6} \cdot \text{tr}(\hat{\Sigma})/N, 10^{-8})$ to guard against ill-conditioning and solves the regularised systems $(\hat{\Sigma} + \varepsilon I)x = \hat{\mu}$ and $(\hat{\Sigma} + \varepsilon I)x = \iota$ with `np.linalg.solve` rather than forming an explicit inverse, (2) computes the unconstrained mean-variance portfolio

$$\kappa = \frac{\iota' \hat{\Sigma}^{-1} \hat{\mu} - \gamma}{\iota' \hat{\Sigma}^{-1} \iota}, \quad \omega^{MV} = \frac{1}{\gamma} \hat{\Sigma}^{-1} (\hat{\mu} - \kappa \iota).$$

and (3) returns the convex combination derived in 2a,

$$\omega_{t+1}^* = \alpha \omega^{MV} + (1 - \alpha) \omega_{t+}, \quad \alpha = \frac{\gamma}{\gamma + 2\lambda},$$

with $\gamma = 4$ fixed throughout. By construction $\iota' \omega^{MV} = \iota' \omega_{t+} = 1$, so $\iota' \omega_{t+1}^* = 1$ holds exactly (to numerical tolerance) without a separate renormalization.

Exercise 3: Covariance Matrix Estimation

We estimate $\hat{\Sigma}_t$ with a Fama–French three-factor model over a rolling 60-month window. For each stock i with at least 36 non-missing observations in the window, we fit

$$r_{i,t} = \alpha_i + \beta_{i,1} f_{MKT,t} + \beta_{i,2} f_{SMB,t} + \beta_{i,3} f_{HML,t} + \varepsilon_{i,t}$$

by OLS. Collecting factor loadings in the $N \times 3$ matrix B , the covariance estimate is

$$\hat{\Sigma}_t = B \hat{F}_t B^\top + \hat{D}_t,$$

where $\hat{F}_t$ is the 3×3 sample covariance of factor returns and $\hat{D}_t = \text{diag}(\hat{\sigma}_{\varepsilon,1}^2, \dots, \hat{\sigma}_{\varepsilon,N}^2)$. Imposing factor structure reduces estimation noise in large panels and guarantees a positive-definite matrix as long as all idiosyncratic variances are strictly positive.

3a. Ledoit-Wolf shrinkage on top of the factor model. Following the assignment’s instruction to apply LW shrinkage *on top of* the baseline estimator, we form

$$\hat{\Sigma}_t^{LW} = (1 - \hat{\alpha}_t) \hat{\Sigma}_t + \hat{\alpha}_t \bar{\sigma}_t^2 I_N,$$

where $\hat{\alpha}_t \in [0, 1]$ is the [Ledoit and Wolf \(2004\)](#) shrinkage intensity (scikit-learn’s `LedoitWolf`) and $\bar{\sigma}_t^2 = \frac{1}{N} \text{tr}(\hat{\Sigma}_t)$ is the average diagonal variance defining the identity target. `LedoitWolf` derives $\hat{\alpha}_t$ for shrinking the *sample* covariance, so applying it to the factor-model baseline is a pragmatic hybrid rather than the LW-optimal intensity for this target. Still, $\hat{\Sigma}_t^{LW}$ inherits positive definiteness

from $\hat{\Sigma}_t$ for any N/T , keeping the linear solves stable. We estimate $\hat{\alpha}_t$ only from stocks with a *complete* 60-month history rather than zero-filling, which would deflate partial-history variance and bias the intensity. Since the large majority of universe stocks have a complete window in any given month, this costs essentially no information.

3b. Comparison of $\hat{\Sigma}_t$ and $\hat{\Sigma}_t^{LW}$.

Table 2: Global minimum-variance weights, January 2017: factor model vs. Ledoit-Wolf.

Statistic	Value
Mean weight (factor model)	0.00685
Mean weight (Ledoit-Wolf)	0.00685
Std dev of weights (factor model)	0.01902
Std dev of weights (Ledoit-Wolf)	0.01599
Mean absolute deviation	0.00308
Correlation of weight vectors	0.9727

Table 2 compares the GMV weights implied by $\hat{\Sigma}_t$ and $\hat{\Sigma}_t^{LW}$ for January 2017. Because $\hat{\Sigma}_t^{LW}$ is a convex combination of $\hat{\Sigma}_t$ and a scaled identity, the two weight vectors are highly correlated ($\rho \approx 0.97$), and the extra identity shrinkage pulls weights toward equal-weight, lowering their cross-sectional dispersion. The mean absolute deviation of 0.0031 is economically meaningful but not dramatic, as the factor model already regularises substantially on its own. This is not a one-month artefact: averaged across all 180 backtest months the correlation is 0.978, the mean absolute deviation 0.0029, and the dispersion reduction 0.0027, matching the January 2017 snapshot.

Exercise 4: Expected Return Estimation

We adopt a hybrid expected-return estimator to cover the full January 2010 – December 2024 window.

January 2010 – December 2021 (primary). We use the Elastic Net model from MA2: at each month t we rebuild the same 60-column feature vector used in training (ten firm characteristics, ten SIC industry dummies, and forty characteristic-by-macro interaction terms, the four macro series entering only through these interactions), apply the fitted `StandardScaler`, and produce predictions $\hat{\mu}_t$, with missing characteristics imputed by the cross-sectional median. The fitted model is very sparse, with only 3 non-zero coefficients out of 60, reinforcing that the cross-sectional signal it captures is weak.

January 2022 – December 2024 (fallback). Characteristics data ends in December 2021, so for the remaining 36 months we use the 60-month rolling historical mean excess return as $\hat{\mu}_t$ (the assignment’s explicit alternative), using only data up to t . Its raw cross-sectional dispersion is dominated by estimation noise rather than signal, so we shrink each stock’s mean toward the cross-sectional grand mean $\bar{\mu}$ with James-Stein empirical-Bayes shrinkage:

$$\hat{\mu}_i = \bar{\mu} + w_{JS}(\hat{\mu}_i^{raw} - \bar{\mu}), \quad w_{JS} = \max\left(0, 1 - \frac{(N-3)\widehat{se}^2}{\sum_i (\hat{\mu}_i^{raw} - \bar{\mu})^2}\right), \quad \widehat{se}_i = \hat{\sigma}_i / \sqrt{T_i},$$

re-estimated fresh each month. $w_{JS} \in [0, 1]$ measures how much of the cross-sectional spread is distinguishable from noise, and empirically it is at or near zero in nearly every fallback month (quantified in Exercise 5), so the TC-adjusted strategies are driven almost entirely by $\hat{\Sigma}_t$ then. The switch is automatic in the backtest loop: when fewer than ten universe stocks have characteristics at month t , $\hat{\mu}_t$ falls back to the rolling mean for that month.

Exercise 5: Portfolio Backtest

The evaluation window is the fixed out-of-sample period January 2010 to December 2024. At each month t we set $\lambda = 0.1$ (1000 bp) and $\gamma = 4$. The expected return $\hat{\mu}_t$ comes from the Elastic Net through December 2021 and the 60-month rolling historical mean thereafter, as in Exercise 4.

Two distinct cost objects appear, and we keep them separate. *Inside* the optimization, the assignment's quadratic penalty $\lambda(\omega - \omega_{t+})' \hat{\Sigma}_t (\omega - \omega_{t+})$ with $\lambda = 0.1$ governs how far the portfolio rebalances. It is a penalty in the objective, not a realized cost (Exercise 2b). *Separately*, for net-performance evaluation we apply an ex-post linear implementation cost following Hautsch and Voigt (2019): with pre-rebalancing weight ω_{t+} and rebalancing distance $\Delta_t := \omega_{t+1} - \omega_{t+}$,

$$r_{t+1}^{net} = \omega_{t+1}' r_{t+1}^{excess} - \beta \|\Delta_t\|_1, \quad \beta = 50 \text{ bp}, \quad \text{TO} = \frac{1}{T} \sum_t \|\Delta_t\|_1,$$

matching equation 46 of Hautsch and Voigt (2019): $\text{TO}_t = \|\Delta_t\|_1$ is the total absolute change in weights (the standard L1 turnover of DeMiguel et al. (2009)), and $\beta \|\Delta_t\|_1$ is the proportional cost of that trading, applied consistently to every strategy. Here $\beta = 50$ bp is a standard per-unit cost estimate for liquid US equities (DeMiguel et al., 2009). Only the two TC-adjusted strategies (b, c) are optimized against the quadratic penalty, but all four are *scored* on this stricter linear cost, the economically meaningful one. Δ_t is taken over the union of last and current holdings, so an exiting stock counts as a full sale and an entering one as a full purchase, rather than being dropped when renormalizing to the current universe (which would understate turnover). For the roughly 0.7% of held positions with no return the next month (typically a delisting) we drop them and reinvest pro-rata.

Strategy (d) is the plain short-sale-constrained mean-variance problem $\max_{\omega} \omega' \hat{\mu}_t - \frac{\gamma}{2} \omega' \hat{\Sigma}_t^{LW} \omega$ s.t. $\omega \geq 0$, $\iota' \omega = 1$, with no transaction-cost term in the objective. Following Jagannathan and Ma (2003), whose constrained framework contains no such term, we keep (d) as a clean non-negativity benchmark. Imposing the transaction-cost penalty under the same constraint would be an equally defensible alternative. We solve it numerically by SLSQP. Net returns and turnover for (d) still use the same Δ_t and β , so all four strategies stay comparable net of cost.

Table 3: Out-of-sample portfolio performance, 2010–2024 (net of 50 bp transaction costs).

Strategy	Ann. Mean (%)	Ann. Sharpe	Avg Monthly TO
Naive 1/N	12.0900	0.8280	0.1010
TC-MV (factor model)	4.5400	0.3430	0.5570
TC-MV (Ledoit-Wolf)	5.6700	0.4500	0.4590
No-short MV (Ledoit-Wolf)	8.6600	0.7760	0.1980

Discussion. Table 3 reports the three metrics for all four strategies over the 179 evaluated months.

Which strategy performs best? As shown in Table 3, naive 1/N achieves the highest annualised mean (12.09%) and Sharpe (0.828) at the lowest turnover (10.1%), consistent with DeMiguel et al. (2009): equal weighting is hard to beat when return estimates are noisy. The no-short MV strategy (the plain Jagannathan-Ma problem, no TC term) is a close second (8.66% mean, 0.776 Sharpe, 19.8% turnover), while both unconstrained TC-MV strategies trail badly (4.54% and 5.67% mean, turnover 56% and 46%). The Elastic Net’s weak predictions (only 3 of 60 coefficients non-zero) produce small but unstable cross-sectional tilts, and the TC penalty (pulling weights only 4.8% toward ω_{t+}) cannot offset the resulting churn. This echoes Jagannathan and Ma (2003): a short-sale constraint is an effective implicit shrinkage device, here substituting for an explicit transaction-cost penalty.

Does the covariance estimator matter? Yes, mainly through stability rather than risk modelling: in Table 3, TC-MV with Ledoit-Wolf has both higher mean return and lower turnover (46% vs. 56%) than the factor-model variant, despite identical $\hat{\mu}_t$. The extra shrinkage toward identity dampens the factor model’s more extreme inverse-implied weights, reducing rebalancing and cost drag.

Table 4: Universe-size robustness. Each cell reports the net-of-cost annualized Sharpe ratio, with average monthly turnover in parentheses, at universe caps $N = 150$ (baseline), 300, and 500.

Strategy	$N = 150$	$N = 300$	$N = 500$
Naive 1/N	0.83 (0.10)	0.82 (0.10)	0.78 (0.10)
TC-MV (FF3)	0.34 (0.56)	0.12 (0.70)	-0.05 (1.00)
TC-MV (LW)	0.45 (0.46)	0.22 (0.56)	0.04 (0.77)
No-short (LW)	0.78 (0.20)	0.65 (0.21)	0.60 (0.24)

Is the result robust to the universe cap? Table 4 re-runs the entire backtest at $N = 150, 300, 500$. The ranking is invariant: 1/N and the no-short portfolio stay strong at every cap, while the unconstrained TC-MV strategies decay as N grows, the factor-model Sharpe falling to -0.05 at $N = 500$ as its turnover climbs to 1.00. The cap is thus a tractability choice that, if anything, flatters the optimized strategies rather than manufacturing the result.

Finally, Figure 2 plots the four cumulative net-return paths.

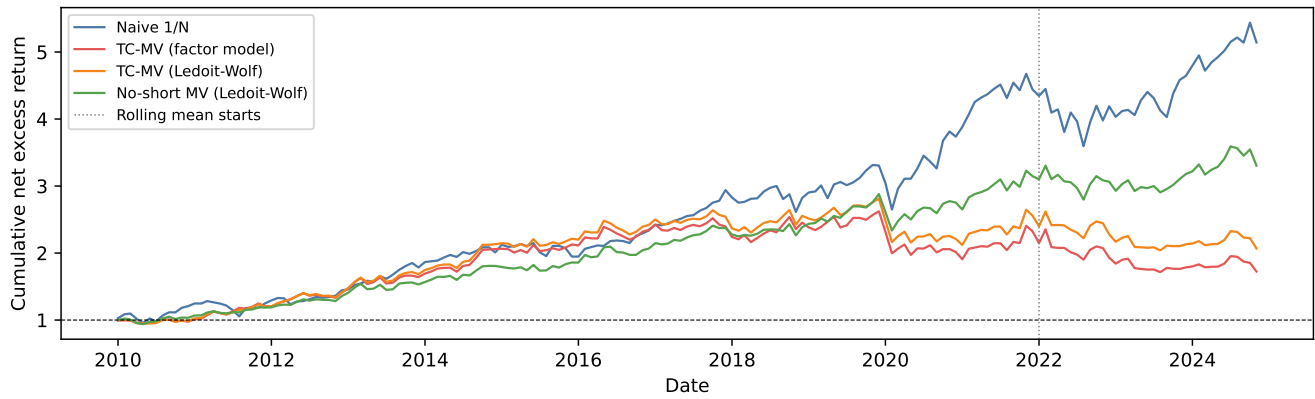


Figure 2: Cumulative net excess return of the four strategies, 2010–2024. The dotted line marks the January 2022 switch from the MA2 Elastic Net to the rolling-mean expected return.

Regime shift at January 2022. As Figure 2 shows, every strategy deteriorates after this date, where $\hat{\mu}_t$ switches from the Elastic Net to the James-Stein rolling mean, including 1/N (13.5% to 6.4% annualised), consistent with a tougher 2022–2024 market (rate hikes, the 2022 sell-off) rather than a methodology artefact. The unconstrained TC-MV strategies turn negative (factor model 7.9% to -9.1%, Ledoit-Wolf 8.6% to -6.4%) with turnover rising sharply (factor model 44% to 103%). Since the average James-Stein weight over all 36 fallback months is only 0.013 (essentially zero), $\hat{\mu}_t$ is near-constant then, so this is not a noisy- $\hat{\mu}_t$ effect: the unconstrained portfolios are driven by $\hat{\Sigma}_t$, whose rolling estimate is simply less stable in this volatile period. The no-short constraint dampens this (turnover only 19% to 24%, mean stays positive), again acting as a stabilising device regardless of covariance estimator.

True out-of-sample? The backtest is out-of-sample conditional on the pre-specified MA2 return signal: at each month t , the covariance estimates, pre-rebalancing weights, transaction costs, and realized returns use only information available at the decision date. The expected-return input is the Elastic Net inherited from MA2, as the assignment requires, so the forecasting model is held fixed rather than re-selected during the backtest. Strictly, this is not a fully nested real-time machine-learning experiment, because the MA2 model was estimated in a previous assignment whose training period partly overlaps the early evaluation window. We therefore read the results as an out-of-sample portfolio evaluation conditional on the MA2 signal, rather than as a fully real-time recursive forecasting experiment.

Exercise 6: Guest Lecture Reflection

Our main takeaway from Cilie Feldager’s guest lecture is that evaluating an AI system is never one-size-fits-all: what counts as good performance depends on the specific use case and the users interacting with the system. Feldager framed this through a “trust contract” (an implicit agreement about what the system promises and what the user can expect), so evaluation criteria must be chosen case by case rather than from a generic checklist. The practical implication is to be explicit, in both design and communication, about what the system can and cannot do for the audience it serves.

References

- DeMiguel, V., Garlappi, L., and Uppal, R. (2009). Optimal versus naive diversification: How inefficient is the $1/n$ portfolio strategy? *The Review of Financial Studies*, 22(5):1915–1953.
- Fama, E. F. and French, K. R. (2008). Dissecting anomalies. *The Journal of Finance*, 63(4):1653–1678.
- Hautsch, N. and Voigt, S. (2019). Large-scale portfolio allocation under transaction costs and model uncertainty. *Journal of Econometrics*, 212(1):221–240.
- Jagannathan, R. and Ma, T. (2003). Risk reduction in large portfolios: Why imposing the wrong constraints helps. *The Journal of Finance*, 58(4):1651–1684.
- Ledoit, O. and Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411.